Fiscal Year:	FY 2017	**Task Last Updated:**	FY 11/10/2016
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Project Title:	Exosomes and Secretory Factors as Mediators of Non-targeted Effects of HZE Particles		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:			
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	This is a proposal devoted to understanding the mechanism of radiation-induced non-targeted effects, and in particular, how they lead to genomic instability in bystander cells (those not directly traversed by a radiation track). Once a bystander cell has become unstable, it is susceptible to mutations induced by endogenous and exogenous stressors, which in turn creates a permissive environment for acquisition of oncogenic mutations. Nontargeted effects (NTE) are a crucial mechanism contributing to solid tumor carcinogenesis in the space environment, yet in many respects they remain poorly understood. We hypothesize that NTE signaling leads to reliance on low-fidelity repair pathways in bystander cells; that the NTE signal is transmitted by exosomes, soluble secretory proteins, or both; and that the NTE signal affects repair through changes in the proteome and transcriptome. To test these hypotheses, we propose three specific aims.		

Task Description:	Aim 1 is to determine effects of radiation quality, dose, and dose rate on genomic instability in bystander cells, as measured by use of low-fidelity repair pathways. Experiments will focus on range of energies for intermediate Z particles. Experiments will use human bronchial epithelial cells, believed to be the target for lung carcinogenesis and bladder cancer cells, used as a positive control because they are known to generate strong non-targeted effect signals. Aim 2 is to determine the non-targeted effect signaling mechanism, with a focus on exosomes and secreted soluble proteins as candidates. Aim 3 is to investigate the exact mechanism by which NTE signaling produces genome instability in bystander cells. In summary, work will improve our quantitative understanding of radiation-induced genome instability, a major mechanism contributing to solid cancer risk, and specifically determine how genomic instability arises as a nontargeted effect in cells not directly traversed by a radiation track.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a project to characterize non-targeted effects of space radiation, that is, effects that are experienced by cells that are not directly traversed by a radiation track. In particular, the project seeks to characterize an effect of space radiation on the DNA repair machinery. When cells that have been exposed to space radiation are allowed to recover, then challenged by introduction of new DNA damage, they have an increased likelihood of repairing that damage in a way that creates chromosomal rearrangements. The effect is seen not only in cells that have been directly irradiated, but is also communicated to their non-irradiated neighbors. This non-targeted effect amplifies the risk of the original exposure. It is important to know how the effects are transmitted and whether transmission can be blocked, perhaps by pharmacological interventions. Prior studies of this phenomenon used genes that had been artificially introduced into the target cells, which created a fluorescent signal when they underwent rearrangement. In a major step forward, more advanced technology has been used to demonstrate that the same effect occurs with endogenous cancer genes in human bronchial epithelial cells, which are thought to be the relevant target cells in lung carcinogenesis. Experiments have now been performed with several different types of space radiation, in order to represent the complex environment in a shielded space habitat. One of the novel aspects of the project is to investigate whether exosomes transmit this non-targeted effect. Exosomes are nanometer-scale, membrane-bounded vesicles capable of carrying information between cells in the form of proteins, RNA, and DNA. New studies, although preliminary, show that exosomes isolated from irradiated bronchial epithelial cells can transmit the non-targeted effect to their non-irradiated counterparts. It is important to note that radiation non-targeted effects, although harmful in the context of space radiation exposure, can be beneficial in the context of cancer therapy, where they are believed to stimulate the ability of immune cells to attack a patient's tumor. Thus, in addition to bringing about a better understanding of how non-targeted effects increase the risks of space radiation, the work may also provide insights into how to make radiation therapy more effective, particularly for late-stage metastatic cancers that are currently difficult to treat.
Task Progress:	This is a project to examine the role of secreted microvesicles, known as exosomes, in non-targeted effects of high charge and energy (HZE) particle radiation. One of these non-targeted effects involves a decrease in the fidelity of DNA double-strand break (DSB) repair, as measured by a propensity to misrepair new DSBs introduced days or weeks following the initial radiation exposure. The effect is seen both in cells traversed by HZE particles (direct effect) and in co-cultured, radiation-naïve bystanders (non-targeted effect). The effect is relevant to risk in the space radiation environment, where occasional encounters with HZE particles occur against a background of much more frequent encounters with lighter, lower-LET (linear energy transfer) particles. A long-lasting change in the fidelity of DSB repair increases the potential for harm from subsequent low-LET induced DSBs. Prior studies of this phenomenon were performed by measuring rearrangements involving artificial reporter genes. The current project uses new technology based on RNA guided endonucleases (RGENs), also known at CRISPR/Cas9. With this technology, studies can be performed in a lung carcinogenesis-relevant epithelial cell model using endogenous cancer-relevant genes. Cells are irradiated and allowed to recover, and the state of the DSB repair machinery is probed by challenging with a pair of RGENs) that introduce DSBs into two loci located 12.5 Mbp apart on the same chromosome. One DSB is located in the EML4 locus, and the other in the ALK proto-oncogene lcous. Faithful repair regenerates the original chromosome, whereas misjoining of the ends creates an oncogenic EML4/ALK fusion gene. This fusion is the single most frequent chromosomal aberration seen in spontaneous human lung cancers. At the end of year 1 and during year 2, cultured bronchial epithelial cells were irradiated at the NASA Space Radiation Laboratory using 16-O, 28-Si, and 48-Ti nuclei at energies calculated to provide linear energy transfer (LET) values of 100 and 200 keV/micron. These represent a range of species that might be encountered within a shielded crew habitat. Low-LET reference irradiation was performed using 137Cs gamma-rays. At six days post-irradiation, cultures were divided. One portion was challenged by expression of the two RGENs (direct effect), and the other was co-cultured with cells expressing the two RGENs (non-targeted effect). DNA was extracted after 72 h, and the relative number of EML4/ALK junctions in each treatment group was measured by quantitative polymerase chain reaction (qPCR). In the cells that were directly irradiated with 48-Ti and allowed to recover, there was a dose-dependent increase in the frequency of RGEN-induced EML4/ALK junctions. Data were consistent with a linear dose response with an average effect size of 7-fold at 1 Gy. There did not appear to be differences between the two energies tested. In cells that were irradiated with 28-Si, there was also an increase in junctions, but it was not dose dependent. Irradiation with gamma-rays or with 16-O had no effect in this assay. Non-targeted effects on co-cultured cells were also not observed under the conditions used. Results provide important proof of concept: they are the first evidence that a history of HZE irradiation adversely affects the ability to repair subsequent damage to an endogenous proto-oncogene in a carcinogenesis-relevant cell type. The effect is dependent on radiation quality and in some cases on dose. The variability between replicate flasks was considerably higher than in prior work using the fluorescent reporter transgene system, which limited the ability to detect small effect sizes. The variability may be due to the complexity of the RGEN system, which requires transduction of the cells with two different vectors and subsequent assembly of protein and RNA components. Several approaches to reduce this variability have been or will be tested, including establishment of stable cell lines in which exposure to a small-molecule inducer results in a uniform pulse of active RGEN expression without the need for variable transductions of individual cell flasks. These lines are expected to be available by the end of the current project year and may provide the additional sensitivity required to measure effects with lighter ions or in co-cultured cells. Separate experiments were performed to investigate whether small, lipid

	bounded microvesicles, known as exosomes, are able to transmit the non-targeted effect on DNA repair. These vesicles carry DNA, RNA, and proteins between cells. It is known that radiation stimulates their formation and release. Because exosomes can be isolated and concentrated from conditioned medium, they provide a potentially more sensitive way to detect the non-targeted effect than would be seen simply using a co-culture approach. Pilot experiments were performed to optimize the protocol for obtaining active exosomes from irradiated cell conditioned medium. To conserve the HZE-irradiated samples, the pilot experiments were performed using cells exposed to a high dose (10 Gy) of low-LET gamma-rays. In a preliminary experiment, RGEN-expressing bronchial epithelial cells were incubated with exosomes for 48 h. DNA was collected and analyzed to determine the frequency of RGEN-stimulated EML4/ALK fusions. Incubation with exosomes from irradiated cells resulted in a 14-fold increase over the background seen in mock-treated control cells. Incubation with exosomes from non-irradiated cells resulted in a much smaller, 2-fold, increase over background. The experiment provides the first experimental evidence that exosomes from irradiated cells may, indeed, be carriers of the non-targeted effect on DNA repair. Large-scale irradiations of bronchial epithelial cells have now been performed with 16-O, 28-Si, and 48-Ti nuclei at energies calculated to LET values of 100 and 200 keV/micron. To recover sufficient conditioned media for experiments, up to 18 large (T-150) culture flasks were irradiated per dose/ion/energy combination. Conditioned media has been collected and reserved for processing to isolate the exosomes. These will be characterized using both the functional assay (effect on repair of RGEN-induced DSBs) and for genomic and proteomic profiling of the exosome contents.
Bibliography Type:	Description: (Last Updated: 08/06/2025)
Articles in Peer-reviewed Journals	Sridharan DM, Asaithamby A, Blattnig SR, Costes SV, Doetsch PW, Dynan WS, Hahnfeldt P, Hlatky L, Kidane Y, Kronenberg A, Naidu MD, Peterson LE, Plante I, Ponomarev AL, Saha J, Snijders AM, Srinivasan K, Tang J, Werner E, Pluth JM. "Evaluating biomarkers to model cancer risk post cosmic ray exposure." Life Sci Space Res. 2016 Jun;9:19-47. http://dx.doi.org/10.1016/j.lssr.2016.05.004 ; PubMed PMID: 27345199 , Jun-2016